

This pointer

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What?

- Every object in C++ has access to its own address through an important pointer called this pointer.
- The “this” pointer is an implicit parameter to all member functions.
- Therefore, inside a member function or constructor, this may be used to refer to the invoking object.
- So, the “this” pointer holds the address of the current object.

Where?

- It can be used to pass the current object as a parameter to another method.
- It can be used to refer current class instance variable.
- It can be used to declare indexers.
- Note: Friend functions do not have this pointer, because friends are not members of a class. Only member functions including constructors have “this” pointer.
- Note: The “this” pointer is a pointer accessible only within the non-static member functions of a class, struct, or union type. It points to the object for which the member function is called. Static member functions don’t have a “this” pointer.

Syntax

this->member-identifier;

Or

(*this).member;

Example

```
#include<iostream>
using namespace std;
class Rectangle
{
    private:
        int length;
        int breadth;
    public:
        Rectangle(int length = 0, int breadth = 0)
        {
            this->length = length;
            this->breadth = breadth;
        }
        Rectangle& setLength(int a)
        {
            length = a;
            return *this;
        }
        Rectangle& setBreadth(int b)
        {
            breadth = b;
            return *this;
        }
}
```

```
void Display()
{
    cout << "Length = " << length << "
    Breadth = " << breadth << endl;
}

};

int main ()
{
    Rectangle rect(10, 5);

    rect.setLength(20).setBreadth(30);

    rect.Display();
    return 0;
}
```

Example

```
#include<iostream>
using namespace std;
class Test
{
    private:
        int number;
    public:
        void setNumber(int number)
        {
            this->number = number;
        }
}
```

```
void Display()
{
    cout << "Number = " <<
    number << endl;
};

int main ()
{
    Test obj;
    int num = 10;
    obj.setNumber(num);
    obj.Display();
    return 0;
}
```